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Orthopedic surgery systems and devices for impacting implements in bones

Abstract

Systems and devices may include a motor operatively coupled to a driveshaft configured to be driven in a first direction in a first mode for impaction and a second direction in a second mode for retraction. Systems and devices may include an anvil configured to be operatively coupled to an implement. Systems and devices may include an impaction cam operatively coupled to the driveshaft, wherein: the impaction cam, when in an impaction position, is configured to strike the anvil when driven by the driveshaft in the first direction to drive the anvil in an impaction direction. Systems and devices may include a retractor configured to be driven by the retraction cam in the second direction, wherein the retractor is configured to strike the anvil to drive the anvil in a retraction direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the priority benefit of U.S. Provisional Patent Application Ser. No. 63/599,219, filed Nov. 15, 2023, and U.S. Provisional Patent Application Ser. No. 63/663,824, filed Jun. 25, 2024, the contents of each of which are herein incorporated by reference in their entirety.

INCORPORATION BY REFERENCE
(1) All publications and patent applications mentioned in this specification are herein incorporated by reference in their entirety, as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference in its entirety.
TECHNICAL FIELD
(2) This disclosure relates generally to the field of orthopedic surgery (e.g., reconstructive surgery of the hip, knee, shoulder, and ankle), the repair of bone fractures, and more specifically, to the preparation of the bone for placement of implant components and insertion of fixation devices. Described herein are systems and methods for easily, safely, and effectively using powered instrument systems for impacting implantable devices into bone.
BACKGROUND
(3) The frequency of Total Hip Arthroplasty (THA) procedures is expected to grow to 635,000 procedures by the year 2030. The procedure relieves pain and improves motion with little to no down time. THA is usually a safe and effective procedure with few complications. When

complications do arise, they may require revision hip surgery. Revision hip surgery can cost upwards of $73,500, may result in long hospital stays, and serious co-morbidities are possible. Complications occur at a rate of 3.5% and include instability and/or dislocation and, especially, peri-prosthetic femur fractures (i.e., fractures that occur around and, as a result of, the insertion of the broach, implant, or stem).

(4) Additionally, the frequency of Total Shoulder Arthroplasty (TSA) procedures is expected to grow to 173,500 procedures by the year 2025. Like THA, TSA has equivalent complications. When complications arise, dislocation and periprosthetic fractures may also occur.

(5) Moreover, surgeons suffer stress on their wrist, arm, shoulder, and back when employing manual hammers or heavy unbalanced impactors during orthopedic surgeries. The stress can lead to workman compensation issues, rehabilitation time off, and a potential reduction in working longevity. The population of orthopedic surgeons is increasing at a 1.5% annual rate while the number of hip surgeries are increasing at a 4.5% annual rate. Surgeons are therefore forced to do more surgeries, thus leaving them even more prone to stress related injuries.

SUMMARY

(6) In some aspects, the techniques described herein relate to an orthopedic impactor system, including: a motor operatively coupled to a driveshaft, wherein the driveshaft is configured to be driven in a first direction in a first mode for impaction and a second direction in a second mode for retraction; an anvil configured to be operatively coupled to a broach or a stem; an impaction cam operatively coupled to the driveshaft, wherein: the impaction cam, when in an impaction position, is configured to strike the anvil when driven by the driveshaft in the first direction to drive the anvil in an impaction direction, the impaction cam is configured to be retarded when a minimum force is sustained by the anvil, when retarded, the impaction cam is configured to be forced out of contact with the anvil in a first axial direction along a longitudinal axis of the driveshaft, and the impaction cam is configured to be forced back into the impaction position in a second axial direction when the impaction cam rotates past the anvil; a retraction cam operatively coupled the driveshaft, wherein the retraction cam is configured to be driven to spin by the driveshaft in the second direction; and a retractor configured to be driven by the retraction cam in the second direction, wherein the retractor is configured to strike the anvil to drive the anvil in a retraction direction,

(7) In some aspects, the techniques described herein relate to an orthopedic impactor system, including: a motor operatively coupled to a driveshaft, wherein the driveshaft is configured to be driven in a first direction and a second direction; an anvil configured to be operatively coupled to a broach or a stem; and an impaction cam operatively coupled to the driveshaft, wherein: the impaction cam is configured to strike the anvil when driven by the driveshaft to drive the anvil in an impaction direction, impaction cam is configured to be retarded when a minimum force is sustained by the anvil, and retarding the impaction cam is configured to force the impaction cam out of contact with the anvil in a first axial direction along a longitudinal axis of the driveshaft.

(8) In some aspects, the techniques described herein relate to a method of using an impactor on a prepared bone including: activating an impactor for impaction using high-frequency, low force impacts, wherein impaction includes: driving a driveshaft in a first direction to drive an impaction cam operatively coupled to the driveshaft, impacting an anvil with the impaction cam such that the anvil is driven in an impaction direction, when a minimum force is sustained by the anvil, disengaging the impaction cam from the anvil by allowing retardation of the impaction cam with respect to the driveshaft and translating the impaction cam in a second axial direction along a longitudinal axis of the driveshaft until the impaction cam is out of contact with the anvil, and forcing the impaction cam back in a first axial direction to an impaction position for a subsequent impact; broaching a canal of the bone; and installing a stem.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing is a summary, and thus, necessarily limited in detail. The above-mentioned aspects, as well as other aspects, features, and advantages of the present technology are described below in connection with various embodiments, with reference made to the accompanying drawings.

(2) FIG. **1**A illustrates an embodiment of an automatic impactor with a double offset implement handle.

(3) FIG. **1**B illustrates the internal components in a housing of the automatic impactor of FIG. **1**A.

(4) FIG. **2**A illustrates a perspective view of an embodiment of an automatic impactor with a double offset implement handle.

(5) FIG. **2**B illustrates a connection mechanism of an embodiment of an automatic impactor.

(6) FIG. **3**A illustrates an embodiment of an automatic impactor with a substantially linear implement handle.

(7) FIG. **3**B illustrates an embodiment of an automatic impactor with a substantially linear implement handle and an eccentric rotating motor (ERM) for generating large amplitude vibrations.

(8) FIG. **4**A illustrates an embodiment of a substantially linear implement handle.

(9) FIG. **4**B illustrates an embodiment of an implement handle.

(10) FIG. **5** is a block diagram of a method of using an embodiment of an automatic impactor system.

(11) FIG. **6**A illustrates an embodiment of a cam assembly that generates a motive impact force on an anvil.

(12) FIG. **6**B illustrates a perspective view of a cam assembly with an attached gear.

(13) FIG. **6**C illustrates a perspective view of a cam assembly including a conical spring.

(14) FIG. **6**D illustrates an exploded view of the cam assembly of an impaction device.

(15) FIG. **7** illustrates an embodiment of a vibratory mechanism.

(16) FIG. **8** illustrates an embodiment of an implement connection mechanism that provides various degrees of orientation of the implement handle relative to the automatic impactor.

(17) FIG. **9** illustrates an embodiment of a camera with a light source trained on a bone.

(18) FIG. **10**A illustrates an embodiment of mode selector input of an impactor.

(19) FIG. **10**B illustrates the internal mechanisms an embodiment of an impactor mechanism, including a retractor mechanism.

(20) FIG. **10**C illustrates the internal mechanisms an embodiment of an impactor mechanism with the retractor removed.

(21) FIG. **10**D illustrates the internal mechanisms an embodiment of an impactor mechanism with the retractor and retraction cam removed.

(22) FIG. **10**E illustrates a perspective view of an embodiment of an impaction mechanism of an impactor.

(23) FIG. **10**F illustrates a top-view perspective of a retraction mechanism of an embodiment of an impactor.

(24) FIG. **11** illustrates an embodiment of a noise reduction mechanism.

(25) FIG. **12** illustrates an embodiment of a balancer connected to an impactor with a sheath.

(26) FIG. **13**A illustrates an embodiment of a retraction assembly of an impaction device.

(27) FIG. **13**B illustrates a perspective view of the retraction assembly of FIG. **13**A.

(28) FIG. **14** illustrates sound absorbing material coupled to an embodiment of an impaction device.

(29) FIG. **15** illustrates a boot coupled to an embodiment of an impaction device.

(30) FIGS. **16**A-**16**B illustrate an embodiment of an impaction mechanism of an impaction device.

(31) FIG. **17** illustrates an embodiment of an impaction device with a gasket for back force mitigation.

(32) The illustrated embodiments are merely examples and are not intended to limit the disclosure. The schematics are drawn to illustrate features and concepts and are not necessarily drawn to scale.

DETAILED DESCRIPTION

(33) The foregoing is a summary, and thus, necessarily limited in detail. The above-mentioned aspects, as well as other aspects, features, and advantages of the present technology will now be described in connection with various embodiments. The inclusion of the following embodiments is not intended to limit the disclosure to these embodiments, but rather to enable any person skilled in the art to make and use the claimed subject matter. Other embodiments may be utilized, and modifications may be made without departing from the spirit or scope of the subject matter presented herein. Aspects of the disclosure, as described and illustrated herein, can be arranged, combined, modified, and designed in a variety of different formulations, all of which are explicitly contemplated and form part of this disclosure.

(34) Surgeons struggle with controlling the broaching process, for example, during the installation of implants into bone (e.g., the femur, humerus, and tibia). A small variance between swings of a hammer can create a large change in impact force between the swings. Several factors may increase difficulty during the broaching process, for example, the large change in force of manual mallets resulting from little variance in swing or the large amplitude motors of impactors known in the art are often do not include options to control force. As a means of facilitating the broaching process, broach handles are utilized. These are implements that connect impactors to broaches and implants, and function to transmit force. Implement handles with strike plates also function to transmit force from a manual hammer. These implement handles may be made of a large mass of stainless steel (e.g., approximately 9 N (2 lbs. of force)) to facilitate stiffness. A particular stiffness may function to transmit energy more effectively. Unfortunately, a large mass of stainless steel also absorbs some of the energy, therefore, creating wasted energy. Wasted energy causes the surgeon to strike the broach handle with more force, further increasing the stress placed on the wrist, arm, shoulder, and/or back. Finally, the surgeon works to overcome the perpendicular frictional force as the broach or implant is advanced into the bone canal. The frictional force is caused by the interaction of the broach or implant when compressing the cancellus bone.

(35) The systems and methods described herein solve the above problems by providing a surgeon with a way to control the force applied with an ergonomic and balanced impactor. Described herein are technical solutions for preventing bone fractures, providing atraumatic ways of managing already fractured bones, and/or alleviating fatigue and stress on surgeons. Described herein are technical solutions for applying force to orthopedic tools and/or orthopedic implants that reduces the likelihood of fractures and/or improves implant seating.

(36) Orthopedic tool may be used herein to describe broaches, reamers, rasps, osteotomes, files, bone taps, awls, curettes, trial components, and the like.

(37) Orthopedic implant may be used herein to describe stems (e.g., femoral stems, tibial stems, humeral stems, radial stems, ulnar stems, and the like), orthopedic nails, and the like. Implement may be used herein to describe the aforementioned tools and implants.

(38) By applying the force with higher frequency and smaller amplitude than a standard mallet or currently available automatic impactor, the surgeon has a further measure of control. Moreover, by using vibration perpendicularly applied to an implement, the amount of force applied can be reduced by, for example, as much as about 20% to about 60%; about 25% to about 55%; about 30% to about 50%; etc. A high frequency vibration overlaying the impaction contributes to diminishing or eliminating the stick-slip effect that otherwise is present when applying a cyclical low frequency impaction where the advancement of the surgical tool stops after an impaction. The high frequency vibration induces micromotion at the interface between the tissue and the surgical tool thus avoiding the transition of the friction forces to a higher static friction force when the motion stops as opposed to lower gliding friction that is present when there is a relative movement between the surface of the surgical instrument and the surrounding tissue. The stick-slip phenomenon has its

principles in the discontinuity between static and kinetic friction. More precisely, the stick-slip effect occurs if the static coefficient of friction is higher than the dynamic coefficient of friction as well and the slide velocity is slow. In this case, overlaying the small amplitude, high frequency vibration maintains the surgical tool in relative motion and thus not falling into the higher static friction envelope. Said another way, once inertia and static friction are overcome, it is more efficient to keep an orthopedic tool and/or orthopedic implant (i.e., implement **107**) in motion, overcoming kinetic friction. These technical solutions provide significant advantages. For example, less energy is used to overcome the higher static friction, since the implement **107** is in a gliding state. For example, control is enhanced over the impaction process due to reduced impaction force used. The precision of the surgical preparation is enhanced due to the substantially continuous gliding impaction process.
(39) Implementation of impaction overlayed with a high-frequency, low-amplitude vibratory motion can be achieved in several ways as described herein. There exist many beneficial applications of this approach. For example, surgeons struggle with removing femoral stems and tibial and femur nails during an orthopedic revision process, as removing a bone nail can be a difficult process. In addition, tooth extraction or removing a medical implement may be difficult. The systems and methods described herein solve the above problems with the use of vibration singularly or in combination with a mallet, forceps, or an impactor. The reversing mechanism of the impactor also provides sufficient force for extraction in addition to the aforementioned capabilities.
(40) Nailing fractured bones, such as shoulder, arm, femur, tibia, etc., can be problematic because of the difficulty in controlling manual hammers. Installing nails with too much or too little force can create unsatisfactory repairs. The systems and methods described herein solve the above problems by employing the impactor described herein to control force and, in some embodiments, utilize vibration as an aid.
(41) Another use case may be animals. For example, large breed dogs such as Great Danes, Saint Bernard, Labrador Retriever, and German Shepherds can suffer from hip dysplasia, a genetic disease. Small dogs may have bone growth plate fractures during development. Small and miniature horses also can be afflicted with hip problems. Dog owners may employ veterinary surgeons to replace the hip joints or repair growth plates with nails. Implants and nails used in these cases are often small. Therefore, pediatric or small adult implants or small nails are utilized. Veterinary hammers are utilized to fixate these implements. Even though the hammers are smaller than adult versions, small hammers still present control problems.
(42) The systems and methods described herein solve the above problems by employing the impactor described herein. Thus, controlling the force and/or utilizing vibration as an aid in solving this problem.
(43) During orthopedic surgery for implants and repair, a variety of hammers are often employed for chiseling (including an osteotome), gouging, splitting, tampering, and/or slicing. Multiple hammers can be utilized with some dubious results. Using multiple hammers may cause the surgeon to change out several hammers during a surgery. For example, a myriad of hammers may include: Cloward, Narrow Hip, Heath, He-Man, Herzog, Repercussion Free Mallet and including, but not limited to, various weights of hammers. These weights for example can be about 4.5 N (1 lb. of force), about 9 N (2 lbs. of force), about (2.5 lbs. of force), about 13.5 N (3 lbs. of force), etc.
(44) The systems and methods described herein solve the above problems by employing the impactor described herein. The impactors described herein may be scaled appropriately for different applications and uses. Solutions described herein eliminate stocking, sterilizing, and providing a myriad of hammers during surgery. Controlling the force and utilizing vibration to reduce the force used and/or a smaller version help solve this technical problem.
(45) The installation of cages in spinal fusion includes similar problems found in orthopedic implant surgery. Whether it is lateral, posterior, or an anterior approach to the surgery, a large force is used. With the force used, it may be awkward to hammer the cage to fixation.

(46) The systems and methods described herein solve the above problems by employing the impactor embodiments described herein. A large force may not be used because of the use of vibration and a smaller force from the impactor. Further, an offset handle as described herein may provide easier spinal vertebral access to allow for fixation of the spinal cage.
(47) Alveolar bone resorption during tooth extraction can cause trauma and bleeding. The systems and methods described herein solve the above problems by employing a smaller version of the impactor described herein. By utilizing vibration and/or a smaller force, the tooth extraction can be less traumatic. Utilizing impactors described herein may replace additional surgical devices that are conventionally used, for example, elevators, peristomes, and forceps.
(48) There presently exists commercial and residential applications in which hammers, and bulky impactors are not practical or ergonomic. These problems include, but are not limited to, hammering nails in confined spaces, breaking up wood knots, making steel letters more pliable, or in agriculture, where large impact drivers are used to drive metal tipped pensile harrow points into the ground to break up the soil.
(49) The systems and methods described herein solve the above problems by employing a straight or offset version of an implement handle that is connected to the impactor. The handle is lighter but stiff enough to deploy the force. Compared to a standard chisel or implement, the work to use the impactor and connected implement is not as tiring.
(50) The systems and devices described herein function to broach bones (e.g., femurs and humeri) and create sufficient space for the installation of an implant or stem. In some embodiments, the systems and devices function to install a stem and/or acetabular component. In some embodiments, the systems and devices employ vibration to aid in broaching and/or installation of a stem and/or acetabular component. In some embodiments, the systems and devices provide light and/or a camera system to view and/or memorialize the cortical rim area or any bone area. In some embodiments, a lightweight and/or rigid implement handle is employed. The systems and devices are used for orthopedic hip and shoulder implant surgery, but can additionally, or alternatively, be used for any suitable applications, clinical or otherwise as described herein. The systems and devices can be configured and/or adapted to function for any suitable spine, trauma, sports medicine surgery, reconstructive surgery, ENT, veterinary surgery, dental surgery, and commercial and residential uses.
(51) Now referring to FIGS. **1**A-**1**B, in some embodiments, an automatic impactor **100** is shown with a double offset implement handle **106** and implement **107**. As shown in FIGS. **1**A, a longitudinal axis **654** of the implement handle **106** is approximately parallel to a longitudinal axis **660** of the implement **107** but is not colinear with the longitudinal axis **660** of the implement **107**. The implement handle **106** serves to connect and transmit motive force from the impactor to the implement **107**. The implement handle **106** can also serve to transmit perpendicular or, in some cases, in-line vibration to the implement (e.g., broach or stem). The impactor **100** may include a battery **101**, a housing **102**, a control input **103**, variable vibration trigger **104**, a connection mechanism **105** for coupling to an implement handle **106** and implement **107**. In some embodiments, the battery **101** may be operatively coupled such that the battery **101** is quickly coupled and decoupled to facilitate battery **101** changes and/or charging. As shown in FIG. **1**B, the housing **102** may at least partially cover and/or hold a plurality of components. The plurality of components may include an electric motor **108**, a gear train, an impaction cam **114** (shown in FIGS. **6**A-**6**D), a conical spring **113** (shown in FIGS. **6**B-**6**C), an anvil **118**, one or more batteries **101**, an electronic driver **109**, and other electronic control elements (e.g., an H-Bridge, a microcontroller, a polyfused, etc.). The electronic driver **109** may be used for speed control of the motor **108**, DC to AC inversion, and/or control of motor direction. As shown in FIG. **6**B, the gear train may include a ring gear **117***a* driven by a conical pinion **117***b*. Other gear trains or alternatives contemplated herein include chain and sprocket, belt and pulley, etc. In some embodiments, a hermetic chamber **119** (shown in FIG. **1**B) may provide humidity and at least partial temperature

insulation for the motor **108** and electronic components. The electric components could be integrally formed in, or fixed to, a single electronic board (e.g., a printed circuit board).

(52) As shown in FIG. **1**A, impactor **100** may provide controlled percussive impacts in a direction (i.e., impaction direction or retraction direction) along a longitudinal axis **654** (parallel to the longitudinal axis **680** of the anvil **118** shown in FIG. **6**A) of the implement handle **106**. The longitudinal axis **654** of the impactor **100** is approximately parallel to a longitudinal axis **660** of the implement **107**. As such, the percussive impacts may be transferred via the implement handle **106** into the implement **107** along the longitudinal axis **660** of the implement **107**. The percussive impacts can be controlled by using a control input **103**, for example a variable force trigger or a variable position trigger. For example, the control input **103**, when pressed, may complete an electrical circuit and transmit current from the battery **101**, or another power source, to the motor **108**. The harder the trigger is pressed or the further the trigger is moved (i.e., the more the trigger is depressed), the more current flows, allowing for finer control over the frequency of impactor **100**. The control input **103** may control frequency proportionally, for example, a trigger depressed 25% results in an impact frequency that is 25% of the maximum frequency. The control input **103** may perform in discrete steps, for example, position thresholds for low, medium, and high may be used for discrete control of the impactor frequency. The control input **103** may vary impact frequency, for example, from about 0 Hz to about 35 Hz or more. In some embodiments, the frequency range is from about 0 Hz to about 25 Hz. In some embodiments, the frequency range is from about 0 Hz to about 10 Hz; about 0 Hz to about 15 Hz; or about 0 Hz to about 20 Hz. The impactor **100** may transmit wireless communication to the wireless communication electronics of the vibratory assembly **130**, as shown in FIG. **3**B, in the implement handle **106**. The wireless communication between the control input **103** and the wireless electronics **131** serves to attenuate the frequency of the vibrations of the vibratory assembly **130**, as shown in FIG. **7**. In some embodiments, the signal transmitted to the vibratory assembly **130** may include an activation signal. For example, when the control input **103** is pressed and the impactor is active, a signal to activate the vibratory assembly **130** may be transmitted to the vibratory assembly **130** to vibrate at a predefined frequency or in a predefined frequency range. Otherwise, the vibratory assembly **130** may not be active. In some embodiments, the frequency of vibrations produced by the vibratory assembly **130** may be controlled by the control input **103**. For example, positional measurement or current measurement based on control input **103** position may be used to control the frequency of the vibratory assembly **130**. Further, the frequency of the vibratory assembly **130** may be increased or decreased as the control input **103** is pressed harder (i.e., pressure is increased on control input **103**). In accomplishing wireless transmission to the vibratory assembly **130**, the impactor **100** may include wireless communication components (e.g., a transmitter, a transceiver, a microcontroller, etc.). The vibration frequency from the vibration motor **135** can vary from about 0 Hz to about 100 Hz or more. In some embodiments, the vibration frequency is from about 0 Hz to about 75 Hz with the vibration applied in a direction **652** (shown in FIG. **7**) perpendicular to the implement (i.e., perpendicular to the longitudinal axis **660** of the implement **107**). Although in some embodiments, the frequency range can range from about 0 Hz to about 50 Hz; about 0 Hz to about 35 Hz; or about 0 Hz to about 125 Hz. The vibrations could also be applied perpendicular and/or laterally. The connection mechanism **105** is configured to provide two or more positions of implement handle **106** for ergonomic insertion in the bone (e.g., femoral, humeri, etc.) canal or for work on any kind of bone. Therefore, in some embodiments, the implement handle **106** (or implement handle **129** shown in FIG. **4**B) can be rotated relative to the impactor **100** in about 70 degree increments to about 110 degree increments; about 80 degree increments to about 100 degree increments; about 85 degree increments to about 95 degree increments; etc. In other embodiments, the implement handle **106** can be rotated in increments of about 5 degrees to about 45 degrees; about 45 degrees to about 120 degrees, about 120 degrees to about 180 degrees, about 120 degrees, about 180 degrees, or any degree amounts therebetween. The implement handle **106** (or implement

handle **129** shown in FIG. **4**B) is connected to the impactor and implement **107**. The implement handle **106** (or implement handle **129** shown in FIG. **4**B) serves to transmit the impact force to the implement. The implement handle **106** (or implement handle **129** shown in FIG. **4**B) serves to transmit vibrations to the stem and broach.

(53) In FIG. **2**A, an embodiment of a double offset implement handle **106** is shown in a rectangular and tubular configuration. It is contemplated herein that implement handles **106** (**129** shown in FIG. **4**B) may be rectangular (shown), tubular (shown), or any other cross-sectional shape with a moment of inertia that provides a desired amount of rigidity. A connection mechanism **321** can connect implement **107** to the implement handle **106**. For example, a connection mechanism may be coupled to the implement handle **106** using a coupling element. Non-limiting examples of coupling elements may include an over-center toggle latch, a roller lock, a flat face lock, a quick-release coupling (e.g., a push button, twist lock, a collar and ball-detent, etc.), a bayonet coupling, or any other appropriate connection or coupling known in the art. FIG. **2**B illustrates a connection mechanism **421** that may be used for connecting the implement handle **106** to the impactor **100** or the implement handle to a implement **107**. The connection mechanism **421** defines an aperture **426** that may receive an attachment portion **422** of the implement handle **106** complementary to the defined aperture **426**. The connection mechanism **421** may lock the attachment portion **422** to the impactor **100** when the collar **425** slides in direction **432** distally away from the impactor **100**. The collar **425** may be biased toward direction **432**, for example, using an operative coupling to a spring. The collar **425** may be rotated about longitudinal axis **680** (i.e., longitudinal axis of the anvil **118**). The collar **425** may define one or more notches **424** substantially complementary to one or more respective posts **423** fixed to, or integrally formed in, the anvil **118**. In a locked arrangement, the collar **425** may be positioned in direction **432** distally away from the impactor **100** at a locked position, locking the attachment portion **422** within the connection mechanism **421**, and the collar **425** is rotated about longitudinal axis **680** such that the one or more notches **424** do not align with the one or more posts **423**. Impaction force and, in some embodiments, retraction forces are generated along longitudinal axis **680**, causing accelerations and, thus, generating forces upon the collar **425** in the opposing direction **430** proximally toward the impactor **100**. When the collar **425** is forced in direction **430**, which is proximally toward the impactor **100**, the lack of alignment of the one or more notches **424** and the one or more posts **423** may lock the collar **425** in the locked position to avoid travel of the collar **425** in direction **430** proximally toward the impactor **100**. In an unlocked arrangement, the collar **425** is rotated about longitudinal axis **680** such that the one or more notches **424** align with the one or more posts **423**. Once the one or more notches **424** are aligned with the one or more respective posts **423**, the collar **425** has clearance to be forced in direction **430** proximally toward the impactor **100** to an unlocked position, thus releasing the attachment portion **422**. It may be advantageous, especially with embodiments utilizing a double offset implement handle **106**, to adjust the rotational position of the implement handle **106** with respect to longitudinal axis **680**. Some embodiments may include a circular attachment portion **422** and circular defined aperture **426**, such that any rotational position can be achieved. Some embodiments may include a circular attachment portion **422** with a key and a circular defined aperture **426** further defining one or more keyways, complementary to the key of the attachment portion **422**. The number of rotational positions may be directly related to the number of keyways (e.g., one keyway may be one rotational position, two keyways may be two rotational positions, three keyways may be three rotational positions, etc.). Some embodiments may include the opposing arrangement (i.e., a key further defined by the defined aperture **426** and one or more keyways defined by the attachment portion **422**). Some embodiments may include an attachment portion **422** and complementary defined aperture **426** that are symmetric shapes. For example, a rectangle may achieve two rotational positions, an equilateral triangle may achieve three rotational positions, a square may achieve four rotational positions, a pentagon may achieve **5** rotational positions, a hexagon may achieve **6** rotational positions, a heptagon may achieve **7**

rotational positions, an octagon may achieve **8** rotational positions, etc. Some embodiments may include an attachment portion **422** that is splined and a defined aperture **426** includes a complementary spline socket profile. A spline and defined spline socket may include a plurality of rotational positions.

(54) In FIG. **3**A, an illustration of a substantially linear implement handle **129** coupled to an impactor **100** is shown. The embodiment includes an implement handle **129** with a longitudinal axis **654** approximately parallel to a longitudinal axis **660** of the implement **107**. The longitudinal axis **654** of the implement handle **129** and the longitudinal axis **660** of the implement **107** are approximately colinear. In some embodiments, the longitudinal axis **654** and the longitudinal axis **660** may also be approximately parallel and approximately colinear with the longitudinal axis **680** of the anvil **118**, as shown in FIGS. **2**B and **6**A.

(55) In FIG. **3**B, an illustration of a substantially linear implement handle **129** is shown with a vibratory assembly **130** that generates high frequency vibrations. The illustrated embodiment includes vibration mechanism **134** (shown in FIG. **7**) with a brush or brushless vibratory motor **135** (shown in FIG. **7**), wireless communication electronics (e.g., a receiver, a transceiver, a microcontroller, etc.), vibratory motor battery, H-bridge or microcontroller, motor and electronics housing, and/or a manual switch or variable speed control **136**. Alternatively, vibrations can also be generated with a piezoelectric or a mechanical device. The brush or brushless vibratory motor **135** and one or more eccentric weights **142** are shown in FIG. **7**. The motor **135** drives the one or more eccentric weights **142** to rotate about the longitudinal axis **650** of the motor **135**. As such, the vibration mechanism **134** may generate vibrations perpendicular to the longitudinal axis **650**. When arranged as shown in FIG. **3**B, the longitudinal axis **650** of the motor **135** is approximately parallel to the longitudinal axis **654** of the implement handle **129**. As such, vibrations generated by the vibration mechanism **134** may be approximately perpendicular to the longitudinal axis **660** of the implement **107**.

(56) In FIG. **4**A, an illustration of a substantially linear implement handle **129** is shown with concentric elongate bodies or tubular bodies **922**, **137**. For example, elongate body **137** may be at least partially concentrically disposed in tubular body **922**. Elongate body **137** may be used to prevent buckling of tubular body **922** during impact and to maintain stiffness. In some embodiments, tubular body **922** may be a stainless steel, aluminum, carbon steel, etc. tube. In some embodiments, elongate body **137** may be a carbon tube, stainless steel tube, fiberglass tube, Galvorn tube, a composite tube, or the like.

(57) In FIG. **4**B, a cross-sectional view of an embodiment of an implement handle **129** is shown. The implement handle **129** may be a polygonal shape, for example a tubular shape, a rectangular shape, or the like. Tubular body **922** may be covered by a weave **923** and infused with a resin **924**. In some embodiments, the implement handle **129** may be baked to set the resin. Tubular body **922** may be made from or include stainless steel, aluminum, carbon steel, and the like. The weave **923** may be a woven carbon weave. The resin **924** may be an epoxy resin, a parylene resin, a polyurethane resin, a cyanoacrylate resin, a polyamide resin, a polyethylene resin, or the like. In some embodiments, a coating **925** is applied to create a hydrophobic surface to prevent humidity infusion in the resin **924** during sterilization cycles (e.g., using an autoclave or other high pressure and/or high temperature protocol). The coating **925** may also prevent humidity related discoloration issues (e.g., a milky appearance). The weave **923** can be woven with a herringbone weave and the pick count (number of carbon fibers per inch or metric measurement) may increase at the outside bend radius **926** of the tubular body **922**. Other weave patterns can include twill, satin, etc. for improved formability and strength. The tubular body **922** may have a thickness of about 0.25 mm to about 2 mm; about 0.3 mm to about 1.75 mm; about 0.4 mm to about 1.5 mm, etc. The use of tubular body **922**, along with a weave system, can reduce the weight of a standard broach handle **129** by as much as about 50% over a standard stainless steel implement handle. For example, a standard stainless steel implement handle can weigh as much as about 0.91 kg (2 lbs. of force).

Additionally, the combination of materials in the embodiment described herein achieves increased weight to stiffness ratios. The improved weight to stiffness ratios allows for effective force and vibration transmission. Stiffness is a function described as force divided by the amount of bending. Furthermore, the weave **923** and resin **924** may also prevent the tubular body **922** from buckling during impact. A distal plug **126**, along with an attached spud, may be coupled to the distal end of the tubular body **922** by spinning, ultrasonic welding, tungsten inert gas (TIG) or laser welding, or fitted to the distal end of the tubular body **922**. A collar **127** with locking mechanism may be attached over the spud of the distal plug **126**. A defined potting hole **128**, that may be plugged by a plug, may serve as a receptacle for resin **924** dispersion inside the collar **127**. Resin **924** dispersion may be accomplished by flowing resin in through the potting hole **128** to fill the defined volume of the tubular body **922**. Air escaping may exit through a defined exhaust hole **1028** until resin reaches the defined exhaust hole **1028**. An inverse of this described process is also possible in which the resin enters the exhaust hole **1028** and air escapes from the potting hole **128**. The weave **923** may be fitted over the bent tubular body **922** and inside the collar **127**. The tubular body **922** can be bent into a number of configurations including, but not limited to, single offset and double offset as an example. The plug with an attached spud is also fitted to the proximal end of the implement handle **106** as described herein. A collar **127** is also attached to the plug. The collars **127** may have an undercut on the inside surface. Resin **924** is then dispersed through the potting hole(s) **128** and then baked in a high pressure and/or high temperature device (e.g., autoclave). Once set, the resin **924** may serve to lock the weave **923** in place. The weave locks in place because of the undercut on the inside surface of the collars **127**. The weave **923** with resin **924** can have up to five times the strength of stainless steel. The proximal plug is configured to fit to the connection mechanism **105**. As an alternative to the embodiment described herein, elongate body (ies) **137** can be inserted into substantially linear (nonplanar) tubular body **922** sections, and interfaces **138** (e.g., machined stainless steel, diecast stainless steel, etc.) can serve as the bent offsets. The interfaces **138** with integral spuds can be press fitted, bonded, laser, spun, TIG, or ultrasonically welded into the nonplanar sections of the tubular body **922**. The design of the described assembly is such that the outside joint surfaces may be flush with one another.
(58) FIG. **5** shows a block diagram of a method S**498** for using an impactor **100**. The method includes optionally preparing a bone for broaching in block S**500**; preparing an impactor in block S**502**; activating the impactor in block S**504**; optionally broaching the canal in block S**510**; and optionally installing an implement in block S**512**. In some embodiments, the method functions to install an implement in the canal of the bone. In some embodiments, the method, for example, in the absence of blocks S**500**, S**510**, and/or S**512** may be used for impaction of a tool (e.g., a chisel, a file, a tamp, or the like) and/or installment of an implement (e.g., a nail or the like). The method S**498** can be configured and/or adapted to function for any suitable operation.
(59) As shown in FIG. **5**, an embodiment of a method S**498** for using an impactor **100** includes optional block S**500**, which recites preparing a bone for broaching. For example, the surgeon may prepare a bone for broaching in the femur during hip surgery by removing the acctabulum. In some embodiments, the humeral canal during shoulder surgery may be prepared for broaching by removing the humeral head. Preparing a bone for broaching may further include preparing the cut bone surface, and reaming or drilling the femur medullary canal or, in shoulder surgery, the humeral canal.
(60) As shown in FIG. **5**, an embodiment of a method S**498** for using an impactor **100** includes block S**502**, which recites preparing an impactor. The impactor **100** is prepared for surgery by attaching an appropriate implement handle **106** or implement handle **129** (shown in FIGS. **1**A and **3**A) and attaching a succession of small to large implements (e.g., broaches or stems shown in FIG. **2**A). Additionally, the implement handle **106** may be rotationally positioned to an advantageous position about longitudinal axis **680** (as described for FIG. **2**B). An advantageous position may increase access to a working site, leverage for the user, and/or ergonomics.

(61) As shown in FIG. **5**, an embodiment of a method S**498** for using an impactor **100** includes block S**504**, which recites activating the impactor. The impactor **100** may operate with, for example, a 12, 18, 20, or 24-volt DC variable speed battery powered electrical system. The battery **101** provides current or motive power through a controller **110** such as an H-Bridge or equivalent. The controller **110** changes the frequency and thus the speed of the motor **108** when the control input **103** is depressed. One or more polyfuses **111** serves to protect the electrical components from a large current in rush or back electromotive force when the control input **103** is fully depressed or released quickly. In some embodiments, the motor **108** is brushless and autoclavable. Alternatively, a brush motor may be employed. The cam system serves to turn rotary motion into linear motion so that an anvil **118** that is connected to the implement handle **106** (or implement handle **129** shown in FIG. **4**B) imparts the impact force to the attached implement **107**. Activating the impactor **100** may include driving a driveshaft **121** in a first rotation direction **682** (shown in FIG. **6**A) (i.e., counterclockwise) to drive an impaction cam **114** operatively coupled to the driveshaft **121**. Activating the impactor **100** may include impacting an anvil **118** with the impaction cam **114** such that the anvil **118** is driven in an impaction direction **125** distally away from impaction cam **114**. Activating the impactor **100** may include, when a minimum force is sustained by the anvil **118**, disengaging the impaction cam **114** from the anvil **118** by allowing retardation of the impaction cam **114** with respect to the driveshaft **121** and translating the impaction cam **114** in a second axial direction **685** (shown in FIG. **10**E) toward spring **113** along a longitudinal axis **681** of the driveshaft **121** until the lobe is out of contact with the anvil **118**. Activating the impactor **100** may include forcing the impaction cam **114** back in a first axial direction **684** away from spring **113** to an impaction position for a subsequent impact.

(62) As shown in FIG. **5**, an embodiment of a method S**498** for using an impactor **100** includes optional block S**510**, which recites broaching the canal. For example, the surgeon may broach the femur medullary canal and an implement **107** (e.g., a final broach) may be used to prepare the medullary canal for a final fitting and/or impaction of the implement **107**. For example, the surgeon may broach the humeral canal and an implement **107** (e.g., a final broach) may be used to prepare the humeral canal for a final fitting and/or impaction of the implement **107**.

(63) As shown in FIG. **5**, an embodiment of a method for using an impactor **100** includes block S**511**, which recites activating the impactor for retraction. Activating the impactor for retraction may include driving a driveshaft **121** in a second direction **683** (shown in FIG. **6**A) (i.e., clockwise) to drive a retraction cam **202** (shown in FIG. **10**B). Activating the impactor **100** for retraction may include impacting a retractor **208** (shown in FIG. **10**B) using the retraction cam **202** such that the retractor **208** drives the anvil **118** in a retraction direction **225** proximally toward impaction cam **114**.

(64) As shown in FIG. **5**, an embodiment of a method S**498** for using an impactor **100** includes optional block S**512**, which recites installing a stem. The surgeon performs a final fitting and/or impaction of the implement **107**.

(65) FIG. **6**A shows an embodiment of an impaction cam system. The impaction cam **114** may include an eccentric profile with cam lobes **140**. In some embodiments, the motor **108** (shown in FIG. **1**B) drives the gear train to rotate a driveshaft **121** and, in turn, the impaction cam **114** rotates eccentrically. The anvil **118** may be in sliding contact with the cam lobe **140**. When a cam lobe **140** strikes the anvil **118**, a force is imparted on the anvil **118** in direction **125** along a longitudinal axis **680** of the anvil **118**. Direction **125** is directed distally away from the impaction cam **114** along the longitudinal axis **680** toward the implement handle **106** (shown in FIG. **1**A)

(66) FIG. **6**B shows another view of the impaction cam system, including the impaction cam **114**, the gear train, and the conical spring **113**. The gear train of some embodiments include a ring gear **117***a* driven by a conical pinion **117***b*. As the ring gear **117***a* is driven to rotate, the ring rear gear **117***a*, operatively coupled to the driveshaft **121**, drives the driveshaft to rotate. The arrangement of the conical pinion **117***b* operatively coupled to the ring gear **117***a* may be illustrated in FIG. **10**D.

(67) FIG. **6**C shows an embodiment of an impaction cam system. The impaction cam **114** includes two opposing cam lobes **140**, shown in FIG. **6**A. The impaction cam **114** may be driven to rotate in a direction, for example rotation direction **682** (i.e., counterclockwise), by a motor **108** (shown in FIG. **1**B) and gear train, for example a conical pinion **117***b* and ring gear **117***a* (FIG. **6**B). The cam system includes the impaction cam **114** forced upon by a conical spring **113**, which may be preloaded.

(68) FIG. **6**D illustrates the operative coupling of the impaction cam **114** to the driveshaft **121**. The driveshaft **121** may define a plurality of spherical grooves **618** that are complementary to a plurality of respective ball bearings **619**. The spherical grooves **618** are complementary due to the radius of the grooves **618** being approximately equal to the radius of the ball bearings **619**. The impaction cam **114** defines a plurality of sockets **615** that are complementary to the respective ball bearings **619**. The impaction cam **114** operatively couples to the driveshaft **121** with the plurality of ball bearings **619** restrained in the respective grooves **618** and respective sockets **615**. Each spherical groove **618** includes approximately matching profiles and angular spacing from one another. As such, the impaction cam **114** may rotate with respect to the driveshaft **121** along the profiles of the spherical grooves **618**. When the impaction cam **114** is driven to rotate with respect to the driveshaft **121** in a first rotation direction **682** (shown in FIG. **6**A) (i.e., counterclockwise), the impaction cam **114** may be advanced in a first axial direction **684** (shown in FIG. **10**E) away from the spring **113**. The profile of the plurality of grooves **618** advances the impaction cam **114** in direction **684** away from the spring **113**, and may cease at, a position appropriate for striking the anvil **118** (shown in FIGS. **6**A, and **10**A-**10**F). When the impaction cam **114** is driven to rotate with respect to the driveshaft **121** in a second rotation direction **683** (i.e., clockwise) (shown in FIG. **6**A), the impaction cam **114** may be retracted in a second axial direction **685** (shown in FIG. **10**E) toward the spring **113**. The profile of the plurality of grooves **618** retracts the impaction cam **114** in direction **685**, compressing spring **113** (shown in FIGS. **6**C and **10**E) and retracting impaction cam **114** out of contact with the anvil **118** (shown in FIGS. **6**A, and **10**A-**10**F).

(69) The impaction process has the advantage of rapidly striking the anvil **118** with the impaction cam **114** and disengaging contact between the impaction cam **114** and the anvil **118**. When the motor **108** (shown in FIG. **1**B) drives the driveshaft **121** to rotate in rotation direction **682** (shown in FIG. **6**A) (i.e., counterclockwise), the impaction cam **114** rotates with driveshaft **121**, establishing angular kinetic energy. As a cam lobe **140** of the impaction cam **114** strikes the anvil **118**, the angular kinetic energy is transferred from the impaction cam **114** to the anvil in direction **125** along the longitudinal axis **680** of the anvil **118**. When the anvil **118** withstands the force generated by the impaction cam **114** impact, the impaction cam **114** may be driven backwards with respect to the driveshaft **121**. Said another way, the rotation of the impaction cam **114** may be retarded by the impact and rotate in rotation direction **683** (i.e., clockwise) with respect to the driveshaft **121** when the force transferred into the anvil is sustained by the anvil (i.e., the anvil **118** sustains a minimum force). A minimum force to cause rotation of the impaction cam **114** in rotation direction **683** (i.e., clockwise) with respect to the driveshaft **121** may be equal to or greater than impaction forces described herein (i.e., about 330 N (75 lbs. of force) to about 890 N (200 lbs. of force); about 440 N (100 lbs. of force) to about 800 N (180 lbs. of force); about 550 N (125 lbs. of force) to about 775 N (175 lbs. of force); etc.). Rotation of the impaction cam **114** in rotation direction **683** (i.e., clockwise) with respect to the driveshaft **121** causes the impaction cam **114** to translate axially in direction **685** toward spring **113**, compressing spring **113** (e.g., a conical spring) until contact between the cam lobe **140** of the impaction cam **114** and the anvil **118** is ceased. After the cam lobe **140** of the impaction cam **114** slides past the anvil **118**, the spring **113** decompresses and advances the impaction cam **114** back in direction **684**. While advancing the impaction cam **114** in direction **684**, the impaction cam **114** is driven to rotate at a more rapid rate than the driveshaft **121**, thus causing the impaction cam **114** to rotate in rotation direction **682** with respect to the driveshaft **121**. The impaction cam **114** angularly accelerates as it advances in direction **684**

until it moves into the appropriate position to once again strike the anvil **118** and repeat the process. The impaction process may be advantageous in that: the motor **108** (shown in FIG. **1**B) does not stall when the anvil **118** sustains the impaction force of the impaction cam **114**, the rotation rate of the driveshaft **121** is substantially constant, and the process produces high-frequency, low force impacts.

(70) FIG. **8** illustrates an embodiment of a connection mechanism **105**. Although there are many approaches to a connection mechanism as defined herein, the illustrated example employs a ball detent **79** system. To decouple the implement handle **106** (or implement handle **129** shown in FIG. **4**B) from the connection mechanism **105**, the collar **143** is slid in direction **226** (in some embodiments against a bias force such as a spring), disengaging the collar **143** from one or more balls **144**. Disengaging the collar **143** from the one or more balls **144** allows the one or more balls **144** to move away from the longitudinal axis **227** and out of engagement with a detent **145** defined by the attachment portion **422**. At this point, the attachment portion **422** is free to be removed from the connection mechanism **105**. Coupling the attachment portion **422** into the connection mechanism **105** may be done by sliding the collar **143** in direction **226** away from the implement handle **106**, allowing the aforementioned clearance of the one or more balls **144** with respect to the attachment portion **422** as it is slid into place. The collar **143** may then be returned in direction **228** toward the implement handle **106** such that the collar **143** removes the clearance that allow the one or more balls **144** to move away from the longitudinal axis **227**, thus locking the one or more balls **144** in the detent **145**. Some embodiments include a connection mechanism **105** with rotational locking capabilities. For example, the detent **145** may further include one or more defined sockets **245** complementary to the one or more balls **144**. As such, once the one or more balls **144** are engaged in the detent **145**, the attachment portion **422**, and thus the implement handle **106** (or implement handle **129** shown in FIG. **4**B), may be rotated **246** about the longitudinal axis **227** until the one or more balls **144** engage with the one or more respective sockets **245**. The collar **143** may be allowed to advance further in direction **228** toward the implement handle **106** further reducing the clearance allowing the balls **144** to move away from the longitudinal axis **227**. As such, embodiments with two or more balls **144** and two or more respective sockets **245** may achieve two or more respective rotational positions that are lockable about the longitudinal axis **227**.

(71) FIG. **9** shows a camera **147** aligned to record the working space and one or more lights **148** aligned to illuminate the working space. In some embodiments, the camera **147** is of sufficient resolution, for example about 50 PPI (pixels per inch) to about 200 PPI, about 60 PPI to about 180 PPI, about 70 PPI to about 150 PPI, etc. A camera module such as Arduino or equivalent with a 4-to-8-megapixel resolution and sufficient focal length to establish picture clarity may be used. A 1× to 2× magnification may be sufficient to see potential hairline cracks. In some embodiments, higher magnification may reduce the field of view. An about 90 degrees to about 120 degree field of view or a fisheye lens may be sufficient for many cortical rim hairline cracks in femurs, for example. Once the broaching is complete and the stem is installed, the view can be memorialized by employing a wireless transmitter connected to the camera to capture the image on a computing device (e.g., cell phone, computer, tablet, etc.). Further, in some embodiments, a microcontroller or other processor may convert the image to gray scale. Gray scale may provide sharper contrast and help to visualize hairline cracks. A swab containing dilute methylene blue powder or methylene blue liquid may be applied. In some embodiments, the swab is diluted greater than about 50% with about 70% or more alcohol concentration. The alcohol in the methylene blue solution reduces the surface contact angle and wets the surface more easily. The methylene blue solution, as a result, more easily seeks a hairline crack making it visible even with an unaided eye. The methylene blue solution can be wiped away with a second clean or alcohol infused swab. The alcohol concentrations can be varied but can be above about 70%. The dilutions with alcohol can be varied but concentrations above about 50% may be sufficient.

(72) Surgical light may be sufficient. However, the surgical area is often a blind hole. A light

emitting diode (LED) having a wavelength of about 365 nm to about 450 nm, or about 450 nm to about 490 nm may be used to improve contrast.

(73) FIGS. **13**A-**14**B shows an embodiment of a reversing mechanism that facilitates the removal or extraction of broaches, stems, nails or any medical or commercial implement. The vibratory assembly **130**, along with an eccentric weight **142**, can facilitate or completely remove or extract an implement (e.g., broach, stem, nail, medical equipment, commercial equipment, etc.). However, in some embodiments, additional, or reverse, impact force may be used to remove or extract the items listed. A double opposing eccentric cam (i.e., impaction cam **114**) may also be used. Either cam is engaged by shifting a shaft **167** in and out. In this way, the motor **108** can be run in reverse or forward.

(74) FIG. **12** illustrates a tool balancing system **168**. Embodiments of the impactor **100** may weigh approximately 4.5 lbs. (2.04 kg) to 5.5 lbs. (2.49 kg). Even though embodiments described include a lighter weight than other known commercial impactors, use over time, for example, a 30-minute period, may tire a surgeon. The effect of impactor **100** weight can be reduced by employing a tool balancer **169** that may be hung on the surgery room ceiling or on an arm **170** connected to a patient support apparatus (e.g., bed, gurney, etc.). A sterile disposable sheath **172** can include a flexible wire with a hook **173** that connects to the fulcrum of the impactor **100**. A force adjustment knob **174** can be accessed through the sheath **172**.

(75) FIGS. **10**B, **10**F, **13**A and **13**B show an impactor **100** that includes a retraction mechanism **201**. It may be advantageous that the impactor **100** include mechanical retraction capability. The retraction mechanism **201** employs a retraction cam **202** that is engaged when the motor **108** is reversed to rotate the driveshaft **121** in rotation direction **683** (shown in FIG. **6**A) (i.e., clockwise). The retraction cam **202** is operatively coupled to the driveshaft **121** using a one-way bearing **203**. As such, the one-way bearing **203** does not engage when the driveshaft **121** is driven to rotate in rotation direction **682** (shown in FIG. **6**A) (i.e., counterclockwise). The one-way bearing **203** may engage to drive the retraction cam **202** when the driveshaft **121** is driven to rotate in rotation direction **683**. When the retraction cam **202** rotates in rotation direction **683**, the lobes **209** may contact a cleat **235** of a retractor **208** and drive cleat **235** in a direction **225**, parallel and opposite of the drive direction **125** (shown in FIG. **6**A) of the anvil **118**. The anvil **118** includes a retraction bar **318** (shown in FIGS. **10**B-**10**F) which is in interference with a strike portion **308** of the retractor **208**. As such, when the retractor **208** is driven back in direction **225**, the strike portion **308** may contact the retractor **208** to drive the anvil **118** back in direction **225**.

(76) As illustrated in FIG. **10**D, some embodiments may include a retraction switch **330** for indicating one or more conditions for retraction. For example, some embodiments may retract when the anvil **118** is pulled in direction **125** distally away from the impaction cam **114** and out of contact with impaction cam **114**. To indicate that the anvil **118** is pulled far enough in direction **125** that the impaction cam **114** will not contact the anvil **118**, the retraction bar **318** may include a limit bar **332**. When the anvil **118** is pulled far enough in direction **125**, the limit bar **332** may contact a limit lever **331** of the retraction switch **330**, actuating the retraction switch **330**. Some embodiments may not power the motor **108** (shown in FIG. **1**B) in a direction to drive the driveshaft **121** in rotation direction **683** (i.e., counterclockwise) unless the retraction switch **330** is properly actuated. The impactor **100** may facilitate the removal or extraction of implements (e.g., broaches, stems, nails or any medical or commercial implement). For example, the impactor **100** may be pulled back such that the anvil **118** is slid to a forward position along direction **125** distally away from the impaction cam **114**. Once the anvil **118** is slid to the forward position, the retraction switch **330** may be actuated such that the retraction cam **202** is driven to rotate in rotation direction **683**. The lobes **209** of the retraction cam **202** impact the cleat **235** of the retractor **208**, driving the anvil **118** in direction **225** proximally toward the impaction cam **114**. An implement may receive the retraction force transferred through the handle **106**, thus driving the implement toward extraction. In some cases, when an implement is successfully retracted, the anvil **118** is driven in direction **225**

proximally toward the impaction cam **114** such that the limit bar **332** is disengaged from the limit lever **331**, thus de-actuating the retraction switch **330** and automatically ceasing the retraction process. Although a lever arm limit switch is shown and described other limit switch and sensor types are contemplated herein. For example, other limit switches and sensors may be inductive proximity switches, capacitive proximity switches, photoelectric switches, reed switches, hall effect sensors, etc.

(77) The motors described herein may be brush or brushless. Brush motors have brushes which are used to commutate the motor to cause it to spin. The armature of a brush motor has a commutator or brush that allows electricity to flow into the armature. The armature is energized with an electro-magnetic field which may cause the armature to spin. A brushless motor is controlled electronically and by definition does not have any brushes. An embodiment may include a brushless motor because it is longer lasting, more energy efficient, has more power per unit volume, and is potentially sterilizable. The motor **108** can be reversed in several ways. One way to reverse the motor **108**, is to reverse the current to a brush motor **108** by employing a double pole double throw switch or a switch with relays or transistor circuit. Motor controls, for either brush or brushless motors, known in the art may be used in the impactor embodiments described herein. FIG. **10**A illustrates an embodiment of an impactor **100** with a mode selector input **600**. The mode selector input **600** may be pressed in direction **602** (opposite direction **604**) to place the impactor **100** in a first mode, for example, an impaction mode. The mode selector input **600** may be pressed in direction **604** (opposite direction **602**) to place the impactor **100** in a second mode, for example, a retraction mode.

(78) The mechanical resistance of the embodiments described herein is such that it doesn't overcome the stall torque of the motor **108** (shown in FIG. **1**B). In some embodiments, a motor with about 12,000 revolutions per minute (RPM) and a gear ratio of at least 10 to 1 may be used. Motor **108** speeds can also vary from about 6,000 RPM to about 8,000 RPM; about 8,000 RPM to about 10,000 RPM; about 10,000 RPM to about 12,500 RPM; or about 12,500 RPM to about 15,000 RPM and beyond. Gear ratios can also be about 5:1 to 10:1; about 10:1 to 15:1; about 15:1 to 20:1; etc. Through experimentation with an embodiment, it was found that a motor with 18,600 RPM and sufficient stall and running torque may be advantageous for embodiments herein. Higher RPMs and somewhat lower speeds may also work. For example, speeds of about 12,000 RPM to about 25,000 RPM may also be suitable. In a particular embodiment, the motor has about 18,600 RPM and the gear train has an about 10 to 1 reduction ratio, putting the impact frequency at a calculated 30 Hz. The actual impact frequency is less, due to frictional losses. At full speed (about 18,600 RPM), the frequency may be about 20 Hz to about 23 Hz (i.e., impacts per second).

(79) Various studies have shown that excessive noise in the surgical operating theater contributes to hearing loss, increased stress, loss of focus, and reduced quality of teamwork. National Institute for Occupational Safety and Health (NIOSH) has established a guideline for workplace noise safety. In this guideline, the noise levels may be less than 85 dBA over an 8 hour period. Further, the allowable noise exposure time limit decreases by half for every 3 dBA increase in loudness. This equates to 15 minutes of exposure for 100 dBA and 3.7 minutes for 106 dBA. The World Health Organization (WHO) has established an Operating Theatre noise guideline as well. The WHO states that the noise must not increase 30 DBA beyond whispering inaudibly in a library. Roughly this is around 60 dBA. Impactor embodiments, described herein, can exceed a sound level of about 100 dBA to about 105 dBA, however, use of the impactor may not exceed about 3 minutes. The noise may be created by the cam lobe **115** striking the anvil **118**. The difficulty with sound mediation is finding sound absorption materials that are sterilizable. Conventional sound absorption materials are thermoplastic polymers that have softening or melting temperatures below sterilization temperatures. Employing a disposable boot that can be ETO (Ethylene Oxide Sterilization) sterilized, or radiation sterilized, is another approach to sound absorption. The “boot” can also serve several purposes including eliminating sterilization of impactor embodiments and

providing a way to attach sound absorption materials.

(80) FIG. **11** illustrates an embodiment including one or more noise reduction mechanisms. By experiment, it has been shown that some embodiments of the impactor **100** have a noise level of above about 100 dB on the A and C scales. Much of the noise is generated by the drive train but primarily by the anvil **118** engaging the cam lobe **140**. The cam housing **116** reduces the noise, at least partially. As described above, the recommended NIOSH noise exposure limit is 85 dB. Several solutions for reducing noise, for example below 85 dB are contemplated herein. For example, vapor deposition of a coating **161**, for example, carbon, ceramic, and/or high temperature polymer coating (e.g., Teflon) on the anvil **118** may be used to reduce or eliminate the metal-to-metal sound during impaction. Another example of sound damping may be to apply a coat **162** to an inside of the cam housing **116** and/or housing **102** with a meta-metal substrate, a graphene infused material, or another sound damping material. Another possible material may be stainless steel foam formed by sintering stainless steel powder with salt crystals and removing the salt crystals to form the foam. For example, sinter stainless foam may be used to coat the inside of the cam housing **116** and/or housing **102**. A subsequent coat in the inside of the cam housing **116** and/or housing **102** may be a thin foamed polyurethane. The above approaches, or combinations thereof, are envisioned to either capture the sound waves or reflect them so that they are inert.

(81) FIG. **14** shows one or more layers **211** of a sound absorbing media (e.g., one or more layers of SonoLayr) attached to a boot **212** (e.g., thermoplastic urethane, thermoset urethane, etc.). The one or more layers **211** of media may be electrospun although other thin electrospun and/or acoustical absorbing materials can be employed.

(82) FIG. **15** shows an embodiment with a boot **212** that may be removably coupled to an impactor **100**. The boot **212** can be made of soft urethane, silicone, or the like. Alternatively, or additionally, the boot **212** can be made of thermoplastic materials such as polyethylene, polypropylene, and thermoplastic rubber. The boot **212** can be designed to attach to an implement **213** that snaps onto the impactor anvil **118**. The other end of the implement **213** can incorporate a connection mechanism similar to the connection mechanism **105** described elsewhere herein. The boot **212** could be disposable. The boot **212** may be designed with one or more pleats **214** that facilitate sterile fitting or dressing of the boot **212** over the L shaped form of the impactor. The boot **212** could fit over the impactor **100** and be tied securely at the open end. The boot **212** could be sufficiently large enough to maintain sterility as it is fitted over the impactor **100**. Pull tabs **218**, extruded from the boot **212**, also serve as a way to maintain sterile technique because the pull tabs **218** may keep a user's hands away from the impactor **100** during removal and installation of the boot **212**.

(83) The anvil **118** and/or the one or more cam lobes **140** may be coated with a sound deadening material such as a PEEK (polyetheretherketone) film. It has been found that coating the anvil **118** and the cam lobes **115** may achieve approximately 3 dBA sound reduction.

(84) Glass microbubbles ranging in size from about 5 microns to about 100 microns can be mixed in a ratio of about 15% to about 75% with the boot **212** material, for example, urethane, silicone, or the like. The glass microbubbles serve to trap the sound, then bounce and dissipate inside the microbubbles. Another sound damping approach may be to keep an air void between one or more sound damping layers (e.g., SonoLayr) and boot **212**. The air between layers of media facilitates sound damping by allowing the sound waves to bounce back and forth. This can be accomplished with dots of hot melt polyethylene or polypropylene dots between sound damping layers and the boot material. Maintaining an air gap between materials can also be accomplished with other materials and methods such as plastic netting and by molding/forming ridges or dots **219** on the inside of the boot or impactor housing **220**. Other materials can be employed for sound damping.

(85) One experiment demonstrated that with a combination of two layers of SonoLayr and Sorbithane, a weighted urethane material, an 8 dBA reduction in sound was achieved.

(86) Another experiment utilized a combination of two layers of SonoLayr and a coating of 50%

glass microbubble/40 shore A urethane mixture, an about 10 dBA to an about 12 dBA sound reduction was achieved.
(87) As a further improvement, a high temperature (e.g., above 350 degrees Fahrenheit) silicone, thermoset, or thermoplastic can be loaded with glass microbubbles **215** in a ratio of about 5% to about 50%. This mixture can be applied as a liner or over molding to the inside of the housings **102** (shown in FIGS. **1**A and **1**B). Based on experimentation, the sound leakage occurs through the spaces around the control input **103**, in the battery compartment **223**, and the anvil **118** area of the housing **102**. These areas may be sufficiently plugged with sound reduction material. For example, sound damping materials, or combinations thereof, may be arranged with, positioned with, and/or coupled to the tolerances between elements of the impactor, for example, the control input **103**, in the battery compartment, and/or the anvil **118** area of the housing **102**.
(88) Further, increased sound damping may be achieved, for example, greater than about 10 dBA to about 12 dBA sound reduction. Increased sound damping can be accomplished by wrapping the cam housing **116** (shown in FIG. **11**) with high temperature silicone and installing or over molding a liner to thin stainless, high temperature polycarbonate or polyester housings **102**. The sound damping solutions described herein may be used alone or combination.
(89) Previously the use of stainless powder with salt crystals was described as a way to manufacture stainless steel foam. Creating a stainless steel foam can also be accomplished through an MIM (Metal Injection Molding) process, or a PM (Powdered Metal) process. In this process stainless steel powder is mixed with either salt crystals, glass microbubbles, ceramic microbubbles, and/or any other microbubble mixture. In the MIM process, the stainless powder is mixed with resin and microbubbles and/or salt crystals. During the injection molding process, a precursor part is molded with the resin binding the mixture described herein. A secondary sintering process at approximately 2000 degrees Fahrenheit melts and drives off the resin microbubble/salt crystal mixture leaving a pocketed foam. This stainless steel foam can be used for the housings **102** (shown In FIGS. **1**A and **1**B) and/or the cam housings **116** (shown in FIG. **11**). A machining step may be used to establish accurate dimensions for the bosses and standing ribs. The MIM process has varying shrink rates that can reach 20%, thus limiting dimensional accuracy. The stainless steel foam can serve to reduce the weight of the housing(s) and/or be used for sound damping. Although the aforementioned may be a difficult process, there is a processing window that would allow for fabricating these parts.
(90) In yet another approach for the housings **102** the MIM, Powdered Metal, precision die casting and other processes can produce a thin stainless part, for example as little as 0.3 mm (0.017 inch) thickness. Thin stainless parts may be particularly useful in reducing weight and/or combining one or more liners or over molding one or more liners for sound reduction.
(91) FIGS. **16**A and **16**B show another approach to an automatic impactor **400** with a modified eccentric-cam linear actuator **401**. Several massaging systems known in the art employ a similar approach but are not useful for orthopedic surgery. Muscles and tendons, as intended, absorb the back force of a massager. However, the back force in a massager style impactor can stress a surgeon's hand, wrist, and shoulder in orthopedic implant surgery. The back force may be mitigated in a surgical orthopedic impactor. Mitigating back force may be useful to the orthopedic surgeon.
(92) Conventional systems employ linear actuator systems for orthopedic surgery. However, linear actuator systems can be cumbersome and difficult to use. For example, these linear actuator systems may be excessively heavy, unreliable, and inherently create a large unmitigated back force. Additionally, linear actuator systems may also have large part counts that negatively affect product reliability. The weight of these systems is typically unbalanced and as result, stresses a surgeon's wrist and shoulder. The unbalanced weight can cause the surgeon to grip harder, further exacerbating the stress on the surgeon's wrist and shoulder. This stress becomes more acute with older surgeons.
(93) FIGS. **16**A and **16**B shows an automatic impactor **400** in which the working mechanism may

be positioned directly over a surgeon's gripped hand (not shown). In this mechanism, the anvil **402** is in a return state **403**. Further, FIG. **16**B shows the anvil **402** in a forward state. The anvil **402** slides in a bushing **411** for a defined distance of **412**. The defined distance **412** is determined by the length of a stroke of the linear crank **406**. A motor, not shown, has a shaft **404** that is connected to a coupling **407** (shown in FIG. **16**A). The coupling **407** is operatively coupled to an eccentric cam **405** and a linear crank **406**. The linear crank **406** incorporates two platens **413** that sandwich a force mitigating buffer **409**. The force mitigating buffer **409** can have a slight preload to reduce the backlash, lash, or play in the mechanism. The force mitigating buffer **409** can be made of high Shore A (**60** to **100**) high temperature urethane, silicone, or a stainless conical spring and the like. It can be made of two or more layers in which a softer layer (30 to 50 Shore A) and a harder layer, as described above, create an increasing resistance as compression is increased. The compressibility of the force mitigating buffer **409** may absorb energy to uniformly reduce the back force. This force mitigating buffer **409** can also reduce the impact force, however, by employing a BLDC (brushless DC electric motor) motor, which is inherently more powerful, the loss of impact force can be offset once the force mitigating buffer **409** is compressed.
(94) FIG. **17** shows another embodiment of a mechanism to mitigate back force. In this embodiment, the handle **414** is isolated from the body of the impactor **415** with a high temperature gasket **416**. The high temperature gasket **416** can absorb the vibrations and back force. The gasket **416** can be fabricated with high temperature silicone, urethane, or any high temperature elastomer. Thermoplastic elastomers can also be employed as a sterile disposable (i.e., single use). Coupled with the urethane or thermoplastic elastomer boot **212** described elsewhere herein, the gasket **416** can offer further back force isolation by allowing limited, independent movement of the impactor **415** with respect to the handle **414**.
(95) The impaction mechanisms described herein are capable of achieving high frequency impacts. High frequency is described herein as about 10 Hz to about 35 Hz; about 15 Hz to about 30 Hz; about 20 Hz to about 25 Hz; etc. The term low impact force used herein is a relative term and should be understood to mean about 330 N (75 lbs. of force) to about 890 N (200 lbs. of force); about 440 N (100 lbs. of force) to about 800 N (180 lbs. of force); about 550 N (125 lbs. of force) to about 775 N (175 lbs. of force); etc.
(96) An experiment was performed to compare the performance of a mallet versus an impactor producing a lower impact force at a higher frequency. The test involved broaching a medium-sized femur to expand its canal for implant insertion using a progressive series of broaches, beginning with a starter broach and incrementally increasing from sizes #6 to #10, with the final broach being a #10. Measurements were taken with a Mitutoyo Absolute Digimatic Vernier Caliper, calibrated and traceable to the National Bureau of Standards, ensuring high precision. Measurements recorded the distance from the top of the cortical rim to the top of the broach at each step. The objective was to compare insertion equivalency between an automatic impactor and a 2-pound manual mallet. During testing, the mallet applied an impact force of 1.2 kN (282 lbs. of force) to 1.4 kN (311 lbs. of force) at a frequency of up to about 2 Hz, whereas the automatic impactor delivered a peak force of 0.75 kN (167 lbs. of force) at about 23 Hz. For the #10 broach, the automatic impactor achieved final insertion at a reading of 11.38 mm to 11.39 mm from an initial position of 18.15 mm. In comparison, the manual mallet achieved final insertion at 11.25 mm to 11.27 mm from an initial position of 18.25 mm. These results demonstrate relative insertion equivalency between the two methods, despite the difference in impact forces, confirming that the automatic impactor can achieve similar final insertion depths as a manual mallet, providing a reliable alternative for femoral broaching.
EXAMPLES
(97) Example 1. An orthopedic impactor system, comprising: a motor operatively coupled to a driveshaft, wherein the driveshaft is configured to be driven in a first direction and a second direction; an anvil configured to be operatively coupled to an implement handle; and an impaction

cam operatively coupled to the driveshaft, wherein: the impaction cam is configured to strike the anvil when driven by the driveshaft to drive the anvil in an impaction direction, impaction cam is configured to be retarded when a minimum force is sustained by the anvil, and retarding the impaction cam is configured to force the impaction cam out of contact with the anvil in a first axial direction along a longitudinal axis of the driveshaft.
(98) Example 2. The orthopedic impactor system of example 1, further comprising a spring configured to force the impaction cam in a second axial direction along the longitudinal axis of the driveshaft into contact with the anvil.
(99) Example 3. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, wherein the impaction cam is operatively coupled to the driveshaft via a plurality of ball bearings interfacing with a plurality of respective spherical grooves defined by the driveshaft and a plurality of respective sockets defined by the impaction cam.
(100) Example 4. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, further comprising a speed control input configured to place the impactor in a first mode or second mode.
(101) Example 5. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, further comprising a battery electrically coupled to the motor.
(102) Example 6. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, wherein the anvil is coupled to the implement using a connection mechanism.
(103) Example 7. The orthopedic impactor system of any one of the preceding examples, but particularly example 6, wherein the connection mechanism includes a plurality of lockable positions with respect to a longitudinal axis of the anvil.
(104) Example 8. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, further comprising a vibratory assembly coupled to the implement handle.
(105) Example 9. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, wherein the system is configured to produce high-frequency, low force impacts.
(106) Example 10. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, further comprising one or more lights configured to illuminate a working space.
(107) Example 11. The orthopedic impactor system of any one of the preceding examples, but particularly example 1, further comprising a camera configured to record a working space.
(108) Example 12. An orthopedic impactor system, comprising: a motor operatively coupled to a driveshaft, wherein the driveshaft is configured to be driven in a first direction in a first mode for impaction and a second direction in a second mode for retraction; an anvil configured to be operatively coupled to an implement handle; an impaction cam operatively coupled to the driveshaft, wherein: the impaction cam, when in an impaction position, is configured to strike the anvil when driven by the driveshaft in the first direction to drive the anvil in an impaction direction, the impaction cam is configured to be retarded when a minimum force is sustained by the anvil, when retarded, the impaction cam is configured to be forced out of contact with the anvil in a first axial direction along a longitudinal axis of the driveshaft, and the impaction cam is configured to be forced back into the impaction position in a second axial direction when the impaction cam rotates past the anvil; a retraction cam operatively coupled the driveshaft, wherein the retraction cam is configured to be driven to spin by the driveshaft in the second direction; and a retractor configured to be driven by the retraction cam in the second direction, wherein the retractor is configured to strike the anvil to drive the anvil in a retraction direction.
(109) Example 13. The orthopedic impactor system of example 12, further comprising a spring configured to force the impaction cam in a second axial direction along the longitudinal axis of the driveshaft into contact with the anvil.

(110) Example 14. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, wherein the impaction cam is operatively coupled to the driveshaft via a plurality of ball bearings interfacing with a plurality of respective spherical grooves defined by the driveshaft and a plurality of respective sockets defined by the impaction cam.
(111) Example 15. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, further comprising a speed control input configured to adjust a frequency of impacts.
(112) Example 16. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, further comprising a mode selector input configured to place the impactor in a first mode or second mode.
(113) Example 17. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, further comprising a battery electrically coupled to the motor.
(114) Example 18. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, wherein the anvil is coupled to the implement using a connection mechanism.
(115) Example 19. The orthopedic impactor system of any one of the preceding examples, but particularly example 18, wherein the connection mechanism includes a plurality of lockable positions with respect to a longitudinal axis of the anvil.
(116) Example 20. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, further comprising a vibratory assembly coupled to the implement handle.
(117) Example 21. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, wherein the system is configured to produce high-frequency, low force impacts.
(118) Example 22. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, further comprising a light configured to illuminate a working space.
(119) Example 23. The orthopedic impactor system of any one of the preceding examples, but particularly example 12, further comprising a camera configured to record a working space.
(120) Example 24. A method of using an impactor on a prepared bone comprising: activating an impactor for impaction using high-frequency, low force impacts, wherein impaction includes: driving a driveshaft in a first direction to drive an impaction cam operatively coupled to the driveshaft, impacting an anvil with the impaction cam such that the anvil is driven in an impaction direction, when a minimum force is sustained by the anvil, disengaging the impaction cam from the anvil by allowing retardation of the impaction cam with respect to the driveshaft and translating the impaction cam in a second axial direction along a longitudinal axis of the driveshaft until the impaction cam is out of contact with the anvil, and forcing the impaction cam back in a first axial direction to an impaction position for a subsequent impact; broaching a canal of the bone; and installing a stem.
(121) Example 25. The method of example 24, further comprising activating the impactor for retraction, including: driving the driveshaft in a second direction to drive a retraction cam, and impacting a retractor via the retraction cam such that the retractor drives the anvil in a retraction direction.
(122) Example 26. The method of any one of the preceding examples, but particularly example 24, further comprising preparing a canal.
(123) Example 27. A method of using an impactor on a prepared bone comprising: activating an impactor for impaction using high-frequency, low force impacts, wherein impaction includes: driving a driveshaft in a first direction to drive an impaction cam operatively coupled to the driveshaft, impacting an anvil with the impaction cam such that the anvil is driven in an impaction direction, when a minimum force is sustained by the anvil, disengaging the impaction cam from the anvil by allowing retardation of the impaction cam with respect to the driveshaft and translating the impaction cam in a second axial direction along a longitudinal axis of the driveshaft until the

impaction cam is out of contact with the anvil, and forcing the impaction cam back in a first axial direction to an impaction position for a subsequent impact; and installing an implement.
(124) Example 28. The method of example 27, further comprising activating the impactor for retraction, including: driving the driveshaft in a second direction to drive a retraction cam, and impacting a retractor via the retraction cam such that the retractor drives the anvil in a retraction direction to remove the implement.
(125) References in the specification to “one embodiment,” “an embodiment” “an illustrative embodiment,” “some embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.[0048] As used in the description and claims, the singular form “a”, “an” and “the” include both singular and plural references unless the context clearly dictates otherwise. At times, the claims and disclosure may include terms such as “a plurality,” “one or more,” or “at least one;” however, the absence of such terms is not intended to mean, and should not be interpreted to mean, that a plurality is not conceived.
(126) The term “about” or “approximately,” when used before a numerical designation or range (e.g., to define a length or pressure), indicates approximations which may vary by (+) or (−) 5%, 1% or 0.1%. All numerical ranges provided herein are inclusive of the stated start and end numbers. The term “substantially” indicates mostly (i.e., greater than 50%) or essentially all of a device, substance, or composition.
(127) As used herein, the term “comprising” or “comprises” is intended to mean that the devices, systems, and methods include the recited elements, and may additionally include any other elements. “Consisting essentially of” shall mean that the devices, systems, and methods include the recited elements and exclude other elements of essential significance to the combination for the stated purpose. Thus, a system or method consisting essentially of the elements as defined herein would not exclude other materials, features, or steps that do not materially affect the basic and novel characteristic(s) of the claimed disclosure. “Consisting of” shall mean that the devices, systems, and methods include the recited elements and exclude anything more than a trivial or inconsequential element or step. Embodiments defined by each of these transitional terms are within the scope of this disclosure.
(128) The examples and illustrations included herein show, by way of illustration and not of limitation, specific embodiments in which the subject matter may be practiced. Other embodiments may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. Such embodiments of the inventive subject matter may be referred to herein individually or collectively by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single invention or inventive concept, if more than one is in fact disclosed. Thus, although specific embodiments have been illustrated and described herein, any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description.

Claims

1. An orthopedic impactor, comprising: a motor operatively coupled to a driveshaft, wherein the driveshaft is configured to be driven in a first direction in a first mode for impaction and a second

direction in a second mode for retraction; an anvil configured to be operatively coupled to an implement handle; an impaction cam operatively coupled to the driveshaft, wherein: the impaction cam, when in an impaction position, is configured to strike the anvil when driven by the driveshaft in the first direction to drive the anvil in an impaction direction, the impaction cam is configured to be retarded when a minimum force is sustained by the anvil, when retarded, the impaction cam is configured to be forced out of contact with the anvil in a first axial direction along a longitudinal axis of the driveshaft, and the impaction cam is configured to be forced back into the impaction position in a second axial direction when the impaction cam rotates past the anvil, wherein the impaction direction is in a radial direction to the first axial direction and the second axial direction; a retraction cam operatively coupled the driveshaft, wherein the retraction cam is configured to be driven to spin by the driveshaft in the second direction; and a retractor configured to be driven by the retraction cam in the second direction, wherein the retractor is configured to strike the anvil to drive the anvil in a retraction direction.

2. The orthopedic impactor of claim 1, further comprising a spring configured to force the impaction cam in a second axial direction along the longitudinal axis of the driveshaft into contact with the anvil.

3. The orthopedic impactor of claim 1, wherein the impaction cam is operatively coupled to the driveshaft via a plurality of ball bearings interfacing with a plurality of respective spherical grooves defined by the driveshaft and a plurality of respective sockets defined by the impaction cam.

4. The orthopedic impactor of claim 1, further comprising a speed control input configured to adjust a frequency of impacts.

5. The orthopedic impactor of claim 4, further comprising a mode selector input configured to place the impactor in a first mode or second mode.

6. The orthopedic impactor of claim 1, further comprising a battery electrically coupled to the motor.

7. The orthopedic impactor of claim 1, wherein the anvil is coupled to the implement handle using a connection mechanism.

8. The orthopedic impactor of claim 7, wherein the connection mechanism includes a plurality of lockable positions with respect to a longitudinal axis of the anvil.

9. The orthopedic impactor of claim 1, further comprising a vibratory assembly coupled to the implement handle.

10. The orthopedic impactor of claim 1, wherein the impactor is configured to produce high-frequency, low force impacts.

11. An orthopedic impactor, comprising: a motor operatively coupled to a driveshaft, wherein the driveshaft is configured to be driven in a first direction and a second direction; an anvil configured to be operatively coupled to an implement handle; and an impaction cam operatively coupled to the driveshaft, wherein: the impaction cam, when in an impaction position, is configured to strike the anvil when driven by the driveshaft in the first direction to drive the anvil in an impaction direction, the impaction cam is configured to be retarded when a minimum force is sustained by the anvil, the impaction cam is configured to be forced back into the impaction position in a second axial direction when the impaction cam rotates past the anvil, and retarding the impaction cam is configured to force the impaction cam out of contact with the anvil in a first axial direction along a longitudinal axis of the driveshaft, wherein the impaction direction is in a radial direction to the first axial direction.

12. The orthopedic impactor of claim 11, further comprising a spring configured to force the impaction cam in a second axial direction along the longitudinal axis of the driveshaft into contact with the anvil.

13. The orthopedic impactor of claim 11, wherein the impaction cam is operatively coupled to the driveshaft via a plurality of ball bearings interfacing with a plurality of respective spherical grooves defined by the driveshaft and a plurality of respective sockets defined by the impaction cam.

14. The orthopedic impactor of claim 11, further comprising a speed control input configured to place the impactor in a first mode or second mode.

15. The orthopedic impactor of claim 11, further comprising a battery electrically coupled to the motor.

16. The orthopedic impactor of claim 11, wherein the anvil is coupled to the implement handle using a connection mechanism.

17. The orthopedic impactor of claim 16, wherein the connection mechanism includes a plurality of lockable positions with respect to a longitudinal axis of the anvil.

18. The orthopedic impactor of claim 11, further comprising a vibratory assembly coupled to the implement handle.

19. The orthopedic impactor of claim 11, wherein the system impactor is configured to produce high-frequency, low force impacts.

20. An orthopedic impactor, comprising: a motor operatively coupled to a driveshaft, wherein the driveshaft is configured to be driven in a first direction in a first mode for impaction and a second direction in a second mode for retraction; an anvil configured to be operatively coupled to an implement handle; an impaction cam operatively coupled to the driveshaft, wherein: the impaction cam, when in an impaction position, is configured to strike the anvil when driven by the driveshaft in the first direction to drive the anvil in an impaction direction, the impaction cam is configured to be retarded when a minimum force is sustained by the anvil, when retarded, the impaction cam is configured to be forced out of contact with the anvil in a first axial direction along a longitudinal axis of the driveshaft, and the impaction cam is configured to be forced back into the impaction position in a second axial direction when the impaction cam rotates past the anvil; the impaction cam including a plurality of sockets to interface with the driveshaft; a retraction cam operatively coupled the driveshaft, wherein the retraction cam is configured to be driven to spin by the driveshaft in the second direction; and a retractor configured to be driven by the retraction cam in the second direction, wherein the retractor is configured to strike the anvil to drive the anvil in a retraction direction.